DATA ITEM DESCRIPTION
Title: Reliability Critical Items List

Number: DI-SESS-80685A
AMSC Number: F10047
DTIC Applicable: No
Preparing Activity: 19
Applicable Forms: N/A

Approval Date: 20190709
Limitation: N/A
GIDEP Applicable: No
Project Number: SESS-2018-044

Use/relationship: The Reliability Critical Items List contains a listing of items which can critically impact the reliability of contract end items. The list includes summaries of and references to specific documents defining compensating controls and features. The list is used to evaluate the adequacy and implementation of critical item controls.

The data in the Reliability Critical Items List will identify and control those items that require special attention because of complexity, application of state-of-the-art techniques, anticipated reliability problems, or the impact of potential failures on safety, readiness, and mission success.

This Data Item Description (DID) contains the format, content, and intended use information for the data product resulting from the work task described in the contract Statement of Work. This DID supersedes DI-RELI-80685.

Requirements:

1. Reference documents. None.

2. Format. The Reliability Critical Items List shall be in the contractor's format.

3. Content. The Reliability Critical Items List shall contain the following:

3.1 A list of items whose substitution or removal would degrade the reliability of the equipment, and that meet at least one of the following criteria:

3.1.1 An operational life of less than 10,000 hours

3.1.2 A calendar life of less than a 10-year period

3.1.3 Two (2) times the program mission life

The maximum allowable operating time or cycles of operations, starting with and including initial functional testing, shall be clearly defined, along with the elements of data and computational methods used to derive those data.

3.2 A list of items difficult to procure or manufacture.

3.3 A list of items with an unsatisfactory operating history.

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3.4 A list of new technology items with little reliability data available.

3.5 A list of items with a single source for manufacturing.

3.6 A list of parts exceeding de-rating.

3.7 Identification of each critical item with cross reference information such that it is possible to trace directly back to the Failure Modes and Effects Criticality Analysis (FMECA), drawings, schematics, and hardware.

3.8 A description of the reason or criteria causing the item to be classified as critical.

3.9 Identification of the activity that discovered the critical items, such as the FMECA, test planning, stress analysis, reliability prediction, or risk assessment, and reference(s) to the related applicable documents.

3.10 A summary of the analysis performed, in specific terms, to identify compensating features, including: design (e.g. extra safety margins), control methods (e.g. overstress testing, process controls, special checkout procedures), or other practices incorporated to minimize the likelihood or effect of the critical items failing during the life of the program.

3.11 A description of the formal policy and procedures for critical item control and notifying affected personnel of the essential and critical nature of such items.

3.12 A description of methods and plans for updating the Reliability Critical Items List to provide timely management visibility.

3.13 A description of the method for identification, control, and test of critical items from initial design through final acceptance.

3.14 Items with single point of failure modes shall be listed in a separate section and shall include Critical Application Items (CAI), when applicable, and Critical Safety Items (CSI). The list shall include the following information for each single point failure mode item:

3.14.1 Rationale for not mitigating the critical item or related single point failure mode.
3.14.2 Impact on the mission, including mission degradation, for not mitigating the critical item or related single point failure mode.
3.14.3 Probability of occurrence of the single point failure mode.
3.14.4 Practicality of incorporating redundancy to remove the single point failure mode.
3.14.5 Recommended options for elimination or mitigation of the single point of failure mode.

3.15 Cross references to the applicable Contract Data Requirements List (CDRL) or cross-referenced document. Data available in other documents submitted with or before the Reliability Critical Items List can be cross-referenced rather than duplicated, if savings result and traceability is not degraded.

End of DI-SESS-80685A

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